# COS 202 FRIDAY–MONDAY ASSIGNMENT

Practical Coding Assignment Submission

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| **Faculty:** | Computing |
| **Course Code:** | COS 202 |

# Assignment Questions

### *Question 1*

Create a personal GitHub account (if you do not have one already) and create a repository named COS202 or any related name, or use any existing one that you have. The repository will be used throughout the semester for storing and managing your practical works, assignments, and projects.

### *Question 2*

Using PYTHON programming language, develop an interactive and well-designed screen MATHEMATICAL CALCULATOR (MC) programme containing components such as +, -, /, *, ^, %, C (Clear), OFF etc. (NB: It is a simple calculator, not a complex scientific one).

### *Question 3*

CGPA is not something that should constrain you to visiting AVERS at all times – this is something you can have inside your pocket and carry all around, running and checking it at your will. Using the powerful provision of selection control statements in PYTHON, develop a PERSONAL POCKET CGPA CALCULATOR (PPC).

*Submission note: Upload the complete source codes and documentation (screenshots, outputs, reports, etc.) of solutions 1 & 2 to your GitHub repository, and submit the repository URL to ajayiprogrammer@gmail.com. Printed source and object listings are to be embedded in the manual, submitted during the Wednesday 22nd test.*

# Answers

## Answer to Question 1 – GitHub Repository

A personal GitHub account was created (or an existing one used), and a repository named **COS202** was set up. This repository is used to store and manage practical works, assignments, and projects for the semester. The source codes for the Mathematical Calculator and the Personal Pocket CGPA Calculator below are uploaded to this repository, and the repository URL is submitted to ajayiprogrammer@gmail.com as instructed.

## Answer to Question 2 – Mathematical Calculator (MC)

The program below is a simple, menu-driven calculator written in Python. It supports addition (+), subtraction (–), multiplication (*), division (/), power (^), modulus (%), clearing the screen (C), and switching off the program (OFF).

```
"""
MATHEMATICAL CALCULATOR (MC)
COS202 Assignment - Simple Screen Calculator
Supports: + - / * ^ % , C (Clear), OFF
"""


def add(a, b):
    return a + b


def subtract(a, b):
    return a - b


def multiply(a, b):
    return a * b


def divide(a, b):
    if b == 0:
        return "Error: Division by zero!"
    return a / b


def modulus(a, b):
    if b == 0:
        return "Error: Division by zero!"
    return a % b


def power(a, b):
    return a ** b


def print_banner():
    print("=" * 40)
    print("      MATHEMATICAL CALCULATOR (MC)")
    print("=" * 40)
    print("Operations available:")
    print("  +   Addition")
    print("  -   Subtraction")
    print("  *   Multiplication")
    print("  /   Division")
```

```
    print("  ^   Power (exponent)")
    print("  %   Modulus (remainder)")
    print("  C   Clear screen")
    print("  OFF Exit calculator")
    print("=" * 40)


def clear_screen():
    print("\n" * 30)
    print_banner()


def main():
    print_banner()

    while True:
        choice = input("\nEnter operation (+, -, *, /, ^, %, C, OFF): ").strip()

        if choice.upper() == "OFF":
            print("\nCalculator switching OFF. Goodbye!")
            break

        if choice.upper() == "C":
            clear_screen()
            continue

        if choice not in ["+", "-", "*", "/", "^", "%"]:
            print("Invalid operation. Please choose +, -, *, /, ^, %, C, or OFF.")
            continue

        try:
            num1 = float(input("Enter first number: "))
            num2 = float(input("Enter second number: "))
        except ValueError:
            print("Invalid number entered. Try again.")
            continue

        if choice == "+":
            result = add(num1, num2)
        elif choice == "-":
            result = subtract(num1, num2)
        elif choice == "*":
            result = multiply(num1, num2)
        elif choice == "/":
            result = divide(num1, num2)
        elif choice == "^":
            result = power(num1, num2)
        elif choice == "%":
            result = modulus(num1, num2)

        print(f"\nResult: {num1} {choice} {num2} = {result}")


if __name__ == "__main__":
    main()
```

**Sample run:** The user is shown a menu of operations. On selecting an operator (e.g. '+'), the program prompts for two numbers, performs the calculation, and displays the result. Entering 'C' clears the screen and redisplays the menu. Entering 'OFF' ends the program.

## Answer to Question 3 – Personal Pocket CGPA Calculator (PPC)

The program below uses Python's selection control statements (if / elif / else) to convert each course score into a grade point, accumulate total units and total grade points across all courses entered, and finally compute and classify the CGPA.

```
"""
PERSONAL POCKET CGPA CALCULATOR (PPC)
COS202 Assignment - Uses selection control statements (if/elif/else)
"""


def grade_point(score):
    """Convert a score (0-100) to a grade point using selection statements."""
    if score >= 70:
        return 5, "A"
    elif score >= 60:
        return 4, "B"
    elif score >= 50:
        return 3, "C"
    elif score >= 45:
        return 2, "D"
    elif score >= 40:
        return 1, "E"
    else:
        return 0, "F"


def classify_cgpa(cgpa):
    """Classify CGPA using selection control statements."""
    if cgpa >= 4.50:
        return "First Class"
    elif cgpa >= 3.50:
        return "Second Class Upper"
    elif cgpa >= 2.40:
        return "Second Class Lower"
    elif cgpa >= 1.50:
        return "Third Class"
    elif cgpa >= 1.00:
        return "Pass"
    else:
        return "Fail"


def print_banner():
    print("=" * 45)
    print("     PERSONAL POCKET CGPA CALCULATOR (PPC)")
    print("=" * 45)


def main():
    print_banner()

    total_units = 0
    total_points = 0.0
    course_count = 0

    while True:
        print(f"\n--- Course {course_count + 1} ---")
        course_name = input("Enter course code (or type DONE to finish): ").strip()

        if course_name.upper() == "DONE":
            break

        try:
```

```
            score = float(input(f"Enter score for {course_name} (0-100): "))
            units = int(input(f"Enter credit units for {course_name}: "))
        except ValueError:
            print("Invalid input. Please enter numeric values.")
            continue

        if score < 0 or score > 100:
            print("Score must be between 0 and 100. Try again.")
            continue

        if units <= 0:
            print("Units must be a positive number. Try again.")
            continue

        gp, letter = grade_point(score)
        course_points = gp * units

        total_units += units
        total_points += course_points
        course_count += 1

        print(f"-> {course_name}: Score={score}, Grade={letter}, "
              f"Grade Point={gp}, Units={units}, Points={course_points}")

    if total_units == 0:
        print("\nNo courses entered. CGPA cannot be calculated.")
        return

    cgpa = total_points / total_units
    classification = classify_cgpa(cgpa)

    print("\n" + "=" * 45)
    print("                RESULT SUMMARY")
    print("=" * 45)
    print(f"Total Courses Entered : {course_count}")
    print(f"Total Units           : {total_units}")
    print(f"Total Grade Points    : {total_points}")
    print(f"Your CGPA             : {cgpa:.2f}")
    print(f"Classification        : {classification}")
    print("=" * 45)
    print("Pocket CGPA Calculator - Carry it everywhere!")


if __name__ == "__main__":
    main()
```

**Grading scale used:** 70–100 = A (5 points), 60–69 = B (4 points), 50–59 = C (3 points), 45–49 = D (2 points), 40–44 = E (1 point), below 40 = F (0 points). CGPA is computed as total grade points divided by total units, then classified from First Class down to Fail.